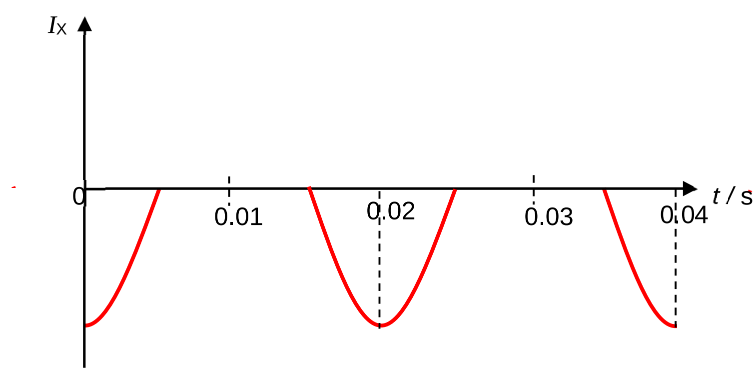
Qns	Answer	Marks
5 (a)	When S is opened, current in coil Y stops flowing and the magnetic flux produced by the current in coil Y decreases. Hence magnetic flux density through **coil X** decreases. By Faraday's Law, an electromotive force is induced in coil X. By Lenz's Law, the induced emf in coil X would be in a direction so as to produce effects to **oppose the decrease** in the magnetic flux density. The induced emf would have produce current, in coil X in the clockwise direction when viewed from Q. But since this current would be in reversed biased direction, LED remain unlit. (For last B1, Student to either describe direction of current to be CW (from Q) leading to LED unlit OR describe current being in reversed biased direction leading to LED unlit.)	**B1** **B1** **B1** **B1**
5b		**B1** Negative cosine graph **B1** Rectified in the correct regions

Qns	Answer	Marks
6(ai)	Lamination increases resistance, thus	M1